

APPROVED METHOD:
OPTICAL AND ELECTRICAL
MEASUREMENT OF ULTRAVIOLET LEDS
AN AMERICAN NATIONAL STANDARD

ANSI/IES/IIUVA LM-92-22

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ULTRAVIOLET LEDS
AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
IES Testing Procedures Committee**



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1.0 Introduction and Scope

1.1 Introduction

This document is a guide developed for the measurement of ultraviolet (UV) light emitting diodes (LEDs), normally in the form of an LED package, array, or module. This document provides uniform methods for operation of UV LEDs and test methods for optical and electrical measurements of UV LED output at a given junction temperature in a continuous-pulse mode, and provides the grounds for specification of UV LEDs at high temperature conditions. The continuous-pulse mode of measurement provides a radiant efficiency (optical power [watts] divided by electrical power [watts]) at a given peak current for a given junction temperature, which allows the calculation of optical power output for a given electrical power input at a specific junction temperature.

The UV optical output of an LED depends on its thermal conditions, in particular, the junction temperature, T_j . Manufacturers of visible-range emitting LEDs normally use single-pulse operation for quality control and binning, whereby LEDs are measured with no heat sink and with the underlying assumption that the junction temperature under these conditions is equal to the room temperature, typically 25 °C. Therefore, published LED specifications are normally for a junction temperature of 25 °C. Furthermore, under a single-pulse mode of operation (often lasting tens of milliseconds) the junction temperature may increase significantly; therefore, the resulting efficiency may correspond to a much hotter junction temperature than the reported 25 °C.

UV LEDs in actual products, however, are typically operating in direct current (DC) and at much higher temperatures (the junction temperature typically ranges from 60 °C to over 100 °C). At these temperatures their emitted flux tends to deviate significantly from what it is under the room temperature condition of 25 °C. Thus, UV product manufacturers often need to know the performance of UV LEDs operating at full rated DC current, and at a thermal equilibrium temperature much higher than 25 °C. To set or measure thermal conditions of the UV LED, case temperature,

pin temperature, board temperature, solder-point temperature, or heat sink temperature is commonly used, depending on the type of UV LED. While these methods are useful to reproduce the same condition for the particular LED, the results using these different methods cannot easily be compared with each other and cannot be reconciled into a universal standard method. Due to the optical characteristics of UV LEDs and their dependence on junction temperature, the only way to obtain reproducible results universally for all types of UV LEDs is by setting them to a specified junction temperature. The test method described in this document sets the UV LED under test to a pre-determined junction temperature for measurement under a continuous-pulse mode of operation. Such a method can establish equivalence of results between the pulse mode operation (normally performed by UV LED manufacturers) and the DC mode operation in products.

The radiometric information typically required for UV LEDs to be used in products includes total radiant flux (watts, or joules per second), centroid wavelength, peak wavelength, and bandwidth. For the purpose of this document, the determination of these values is referred to as *optical measurement*.

The electrical characteristics typically required for UV LEDs to be used in products are peak input current, forward voltage, and input power. For the purpose of this document, the determination of these values is referred to as *electrical measurement*.

1.2 Scope

This document describes the procedures to be followed and precautions to be observed in performing measurements of total radiant flux (total radiant power), electrical power, and wavelength characteristics of ultraviolet (UV) light emitting diodes (LEDs). This document covers measurement of UV LEDs in the wavelength range of 200 nm to 400 nm* under continuous-pulse operation. Where the term “thermal condition” is used with respect to a UV LED, this refers to the junction temperature.

* LEDs with wavelengths longer than 360 nm are covered in ANSI/IES-LM-85-20.¹

For UV LEDS, due to an initial transient voltage effect, DC mode operation and single-pulse operation, as described in *ANSI/IES LM-85-20, Approved Method: Optical and Electrical Measurements of LED Packages and Arrays*,¹ are not acceptable measurement methods.

This document does not cover AC-driven UV LEDs, UV LED lamps, nor UV LED luminaires. This document does not cover relative measurement methods.

This standard does not address the safety concerns associated with the use of these products. It is the responsibility of the user of this standard to establish appropriate safety and health practices.

2.0 Normative References

2.1 ANSI/IES LM-58-20

Illuminating Engineering Society. Approved Method: Spectroradiometric Measurement Methods for Light Sources. New York: IES; 2020.

2.2 ANSI/IES LM-75-19

Illuminating Engineering Society. Approved Method: Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: IES; 2019.

2.3 ANSI/IES LM-78-20

Illuminating Engineering Society. Approved Method: Total Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.

2.4 ANSI/IES LS-1-21

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2021. Online: <https://www.ies.org/standards/definitions/>. (Accessed 2022 Feb 1).

3.0 Definitions

3.1 data acquisition (DAQ)

The process of measuring electrical or physical

phenomena, e.g., current, voltage, temperature, pressure, or sound. The abbreviation DAQ is sometimes used to represent *data acquisition unit*.

3.2 device under test (DUT)

The UV LED that is being tested.

3.3 digital multimeter (DMM)

A device used to measure electrical current, voltage, and resistance. Some types measure other quantities as well.

3.4 junction temperature (T_J)

The temperature of a DUT *p-n* junction; in the case of multiple-junction DUTs, the T_J is considered as a single-junction equivalent that represents the effective (or aggregated) temperature.

3.5 out-of-band light

The light outside the wavelength range that encompasses 97% of the DUT radiant flux.

3.6 pulse period jitter

The maximum peak-to-peak variation in the period of the pulsed current signal, measured from the 50%-positive transition points of two successive pulses. Pulse period jitter may be expressed in microseconds or as a percentage of the period plus a fixed value in microseconds.

3.7 pulse width error

The difference between the actual PCS pulse width, measured from the 50%-positive and 50%-negative transition points, and the nominal pulse width. Pulse width error may be expressed in microseconds or as a percentage of the pulse width plus a fixed value in microseconds.

3.8 pulse width jitter

The maximum peak-to-peak variation in the width of the pulsed current signal, measured from the 50%-positive and 50%-negative transition points. Pulse width jitter may be expressed in microseconds or as a percentage of the pulse width plus a fixed value in microseconds.

3.9 pulsed current source (PCS)

A device that provides the pulsed or DC current that powers the DUT during measurements.

3.10 settling time

The time needed for the current signal to reach a stable level, within 1% of the nominal setting, after a fast transition; it is a function of the PCS, DUT, and cabling.

3.11 source measure unit (SMU)

A type of test equipment that is capable of providing the pulsed current and measuring the forward voltage at the same time.

3.12 temperature-controlled mount (TCM)

A device that incorporates one or more cooling and/or heating components to maintain its temperature at a set value. A TCM is used to control the temperature of an LED mounted on it for testing purposes.

3.13 UV LED

For the purpose of this document, an LED package, array or module emitting in the middle to near ultraviolet range (as defined in ANSI/IES LS-1-21; see **Section 2.4**) that is operated via an auxiliary driver.

4.0 Physical and Environmental Test Conditions

4.1 Thermal Conditions

4.1.1 Junction Temperature, T_J . The optical and electrical characteristics of a DUT vary with its T_J . The T_J of the DUT shall be actively controlled. Reports of optical output of the DUT shall always include the T_J used in the measurement. A T_J of 25 °C is the standard condition; however, elevated T_J , e.g., 50 °C to 100 °C, may also be used to represent the temperature of the DUT operating in real applications.

4.1.2 Temperature Control of the DUT. One of the two methods for DUT temperature control described here shall be used:

- **Using a TCM to control the T_J :** The temperature of the DUT may be controlled using a TCM. The temperature of the DUT is set by a TCM on which the DUT is mounted. For elevated temperatures, T_J shall be controlled using a TCM.

- **Using the ambient temperature to control the T_J :** For the nominal condition of $T_J = 25$ °C, the DUT temperature may be set by the ambient temperature. The T_J of the DUT will increase to unacceptable levels if the heat sink capacity is not sufficient. This method shall not be used for elevated temperature conditions, as a thermal chamber might be required, which could in turn make the total radiant flux measurement difficult. **Annex A** provides additional information.

4.1.3 Ambient Temperature (T_A). Ambient temperature, T_A , may be used to set T_J for the nominal condition of $T_J = 25$ °C (see **Section 4.1.2**). At elevated junction temperature, T_J is controlled by a TCM (see **Section 4.1.2**), and the effect of ambient temperature on the DUT's ultraviolet output will be insignificant only if the thermal resistance from DUT junction to TCM is small. T_A shall be noted in the test report. T_A shall be measured at a point not more than 0.5 m from the DUT and at the same height as the DUT. The temperature sensor shall be shielded from direct radiation of the DUT.

4.2 Airflow

Air movement across the surface of the DUT may alter the T_J . Airflow around the DUT should be such that that airflow does not affect T_J . The amount of airflow that can be tolerated critically depends on the thermal resistance between the DUT and TCM (or DUT mount if T_J is set by T_A).

4.3 Thermal Conditions for Mounting DUTs

For T_J controlled by a TCM, the thermal resistance from the DUT junction to the TCM should be minimized by ensuring good thermal contact (e.g., by applying thermal grease) and the use of materials with high thermal conductivity (e.g., metal-core printed circuit boards, MCPCBs).

4.4 Vibration

There are no specific requirements, but good laboratory practice suggests DUTs should not be subjected to excessive vibration or shock during stabilization, transportation, mounting, or testing.

4.5 Humidity

Relative humidity values greater than approximately 65% can lead to corrosion effects in some instruments, and values below approximately 10% can lead to electrostatic effects. Therefore, laboratory relative humidity should be monitored and maintained between 10% and 65%.

5.0 Electrical Test Equipment

The measurement methods described in this document require instrumentation that can provide current and measure voltage. The pulsed methods require, at a minimum, a calibrated pulsed current source (PCS; see **Section 3.3**) or a pulsed source measure unit (SMU; see **Section 3.8**). For forward voltage (U_F) measurements, a digital multimeter (DMM) or data acquisition (DAQ) instrument is also required. If using the M-DCP optical measurement method (see **Section 8.3**), additional instruments that sense and measure the current pulse train are also required. Finally, an oscilloscope may be useful to visualize the current pulses applied to the DUT to ensure that the measurement setup is operating properly. The required instruments for each task are listed in **Table 5-1**. Detailed instrument requirements follow, in **Sections 5.1** through **5.5**.

5.1 Pulsed Current Source (PCS) or Source Measure Unit (SMU)

The PCS or SMU shall be able to source pulsed constant current at the required amplitude, with programmable pulse width capability in the range of 10 μ s to 50 μ s and programmable pulse frequency capability that supports duty cycles in the range of 1% to 2%. The PCS voltage limit shall be greater than the maximum DUT forward voltage plus any additional voltage drop over electrical connections. The PCS power limitation shall be greater than the maximum required DUT electrical power during the measurement.

The total PCS timing error shall not exceed the width of the short pulse applied as part of the continuous-pulse method. For a typical PCS this is:

$$(\text{positive transition settling time}) + (\text{pulse width error}) + (\text{pulse width jitter}) + (\text{negative transition settling time}) < 10 \mu\text{s}$$

In addition, the PCS period jitter shall not exceed 1% of the period (e.g., less than 10 μ s for a period that is 1 ms).

PCS DC current accuracy shall be assessed to ensure that the current error contribution to the measurement uncertainty is within acceptable limits. In any case the current uncertainty (after settling) shall be less than 0.5%.

Table 5-1. Required Electrical Test and Measurement Equipment

Task	Method			
	DC	DCP or CPT	M-DCP	U_F Measure, I-U Sweep
Source current to DUT	DC current source or SMU	Pulsed current source or SMU		
Measure DUT voltage	Required; may have a large error; for reference only			DMM, DAQ unit, or SMU
Sense DUT current	Not required; relies on the PCS or SMU uncertainty		Current shunt or probe	Not required
Measure sensed current	Not required; relies on the DC current source, the PCS, or the SMU uncertainty		6½-digit DMM or better w/multi-slope integrator	Not required
Visualize current waveform	Not required; if desired, use an oscilloscope or an SMU display			
Control sample temperature	Temperature controlled mount			

Note: Suitable PCS instruments are typically called *precision pulsed current sources* or *pulsed source measure units* (pulsed SMUs). These instruments typically have hardware-based pulse generation capability that supports microsecond pulsing, and they include deterministic hardware-based timing systems that can meet the error and jitter requirements. General-purpose SMUs without hardware-based pulse generation are generally not suitable for accurate measurement.

5.2 Pulse Amplitude Measuring Device

A DAQ instrument, an SMU, or the DAQ function of a DMM may be used for U_F and voltage-current (I-U) sweep measurements. These measurements require the DUT's voltage signal to be averaged over a defined portion of the pulse. The expanded uncertainty of the voltage measurement shall be less than 0.5% ($k = 2$), and bandwidth should be higher than 400 kHz. To ensure that measurements are correctly timed within a pulse, the measurement instrument shall have a timing system with microsecond trigger resolution and a programmable measurement aperture. For lowest noise, it should have a floating, differential input.

Important: This instrument may be used to visualize the overall current waveform, but due to its limited bandwidth and sample rate, it shall not be used to measure or correct the current waveform.

A pulsed SMU that includes a voltage digitizing function is suitable equipment. This arrangement simplifies pulse and measurement coordination. A sampling DMM with programmable trigger and aperture functions may also be used. If a sampling DMM is used, it shall be linked to the PCS with a hardware trigger.

Data acquisition instruments, sometimes called simply DAQs, may also be used to measure DUT U_F . DAQs normally feature fast sampling and excellent trigger resolution; however, they may lack flexibility in measurement aperture programming. If this is the case, samples should be collected and averaged post-measurement. When selecting a DAQ, care should be taken to review its common-mode voltage specifications. To produce fast rise times, some instruments employ amplifiers to apply voltages above the DUT U_F (sometimes 20 to 30 V above). This

voltage is used to overcome cabling inductance. The DAQ common-mode specification should be reviewed prior to selection to ensure that it will not be exceeded.

5.3 Current Shunt or Current Probe

If the M-DCP method (refer to **Section 8.3**) is used, then a current shunt or current probe is required to sense the current through the DUT. These devices sense the current through the DUT and produce an electrical signal that is measured with a DMM or high-speed DAQ. The current shunt or current probe should be attached near the DUT to allow the measurement to account for capacitive cable losses.

Suitable current probes are high-accuracy Hall-effect types. To minimize errors, the probe should be zeroed prior to a measurement, and the probe's amplitude error should be characterized and corrected. Probe bandwidth shall be equal to or greater than 8 MHz.

Suitable current shunts are built into many DMM instruments. External shunts may also be used. These are especially useful for currents above 10 amperes (A). If an external shunt is used, its resistance should be minimized to reduce the additional voltage burden on the PCS or SMU. For example, a 0.1-ohm shunt adds 1 V of extra voltage burden at 10 A. Cable inductance to and from the shunt should be minimized by keeping wires as short as possible. If the shunt is connected to an instrument (DMM, DAQ, or oscilloscope) with an earth ground connection, some DC or AC current may flow back to the PCS through the ground connection, degrading the pulse accuracy. To avoid this, shunts should only be used with isolated instruments, or the shunt's output signal should be conditioned with a differential amplifier prior to connecting it to the grounded instrument.

5.4 DMM with Multi-slope Integrator or High-Speed DAQ

If the M-DCP method (refer to **Section 8.3**) is used, then a 6½-digit DMM or better with a multi-slope integrator is required. These instruments measure the average of the pulsed current signal. This average is then used to correct the measurements.

Commercial benchtop DMM instruments typically have the required multi-slope integrator. This circuit integrates the incoming voltage signal to determine its average amplitude. Because the integrator analog circuitry responds to high-frequency and low-frequency signal components, and because it has excellent accuracy, it is a good choice for measuring the average of the pulse's current signal. DMMs also frequently include suitable current shunts.

Alternatively, a high-speed DAQ may be used to measure the average of the current signal. If a DAQ is used, its sample rate shall be at least 20 million samples per second, and its bandwidth shall be at most 10 MHz.

5.5 Oscilloscope or SMU Display

While not required by the methods described in this document, an oscilloscope may be useful for visualizing the current waveform. Visualization allows the current waveform to be verified, and it is useful when adjusting load tuning settings, available in some PCS instruments. Current waveform visualization may also be achieved using the SMU's digitizing function. If an SMU is used, an external shunt or current probe should be used to minimize errors due to cable capacitance.

5.6 Temperature-Controlled Mount

The measurement methods require the DUT to be maintained at a specific temperature. This is normally done by placing the DUT on a TCM. The TCM shall have a temperature resolution better than 0.05 °C and a stability of 0.1 °C when constant power is applied.

Commercial thermoelectric cooler (TEC)-based mounts are good choices for this function. These mounts may be equipped with mounting attachments that make DUT installation and removal easy. TEC-based mounts should be sized to handle the average power applied to the DUT.

6.0 Optical Measurement and Equipment

For measuring optical parameters, either an integrating sphere or a goniometer shall be used. Requirements for

each system are defined separately, in **Sections 6.1** and **6.2**. The radiometer types and measured quantities for each system are detailed for each of these two types of equipment.

6.1 Integrating Sphere

The integrating sphere of the sphere radiometer shall have a detection port for placing the radiometer and an input port for the DUT, and it should have an auxiliary light source for self-absorption correction. The diameter of each sphere opening should be less than one-third of the sphere's inner diameter. Baffles shall be used ensure that the first ray of light emanating from the source does not contact the auxiliary source, the first ray from the auxiliary source does not contact the source, and neither contacts the detector. (Refer to ANSI/IES LM-78-20 for additional information; see **Section 2.3**.)

6.1.1 Detector. The input optics at the detector port of the integrating sphere (normally equipped with a diffuser) shall have an approximate cosine response. The cosine response of the detector should have a directional response index of less than 15% in the wavelength region of interest.²

The detector should have a capability of measuring pulsed optical radiation accurately. It should be ensured that the detector and measurement circuit do not saturate with a peak detector-output signal that is very high compared to the time-averaged reading. For example, if the duty cycle is 1%, the peak detector current is 100 times higher than the time-averaged reading of the pulse. Normal auto-ranging detectors may not work properly for such pulsed light.

6.1.2 Radiometer. The radiometer is a spectrally integrating detector that may have a spectral filter to limit the wavelength range of the detection (e.g., to avoid detection of ambient radiation or radiation related to sphere fluorescence).

The linearity of the radiometer should be verified. (Refer to *CIE 237:2020, Non-Linearity of Optical Detector Systems* for guidance on testing for linearity.³)

Calibration: The sphere-radiometer system shall be calibrated for total radiant flux traceable to the

International System of units (SI) through a national metrology institute (NMI). The calibration method shall take into consideration the spectral responsivity and geometry (2π or 4π) of the calibration source and method compared to those of the DUT.

Measured quantities:

- **Total radiant flux, Φ_e :** The radiant flux emitted from a source, integrated over all directions (4π sr). Total radiant flux is the output of the sphere-radiometer.
- **Radiant efficiency (of a source):** The ratio of the total radiant flux Φ_e of the emitted radiation to the electrical power P consumed by the source. (Source: ANSI/IES LS-1-21; see **Section 2.4.**) The electrical power P consumed by the DUT shall be measured according to the method described in **Section 8.** The radiant efficiency is given as:

$$\eta = \frac{\Phi_e}{P}, \text{ expressed as a percentage}$$

6.1.3 Spectroradiometer. The spectroradiometer shall cover the wavelength range that is relevant for the application, and the bandwidth (full width at half maximum, FWHM) and scanning interval shall be no greater than 3 nm. Typical systems cover from 200 nm to more than 400 nm. The spectroradiometer system shall have a wavelength uncertainty within 0.5 nm ($k = 2$). The spectroradiometer system should account for light outside of this wavelength range that may result in stray light within the spectroradiometer system, especially during calibration.

The linearity of an array spectroradiometer should be tested for measurement of such continuous pulses. (per ANSI/IES LM-58-20; see **Section 2.1.**)

Calibration: The sphere-radiometer system shall be calibrated for total radiant flux traceable to the International System of units (SI) through a national metrology institute (NMI). One methodology involves the use of a spectral-irradiance standard irradiating a precision aperture. With a source calibrated for W/(m²·nm) irradiating a precision aperture calibrated for m²; the result is measured in watts per nanometer (W/nm) entering the sphere-spectroradiometer. An alternative method uses an external source calibrated

to the SI system of units for relative spectral irradiance for a relative spectral radiant-flux calibration and using an LED calibrated to the SI system of units to set the total radiant flux scale.

Measured quantities:

- **Total spectral radiant flux, $\Phi_e(\lambda)$** (refer to ANSI/IES LS-1-21)
- **Total radiant flux, Φ_e** (refer to ANSI/IES LS-1-21)
- **Centroid wavelength, λ_c** (refer to ANSI/IES LS-1-21)
- **Peak wavelength, λ_p :** Unless specified otherwise, peak wavelength is the wavelength associated with the maximum measured spectral signal. Accurate determination of peak wavelength requires good sampling or interpolation of the measured spectral distribution.
- **Spectral bandwidth, $\Delta\lambda$:** Spectral bandwidth is the full width at half maximum (FWHM) of the total spectral radiant flux distribution, which is the difference between the two wavelengths at which the total spectral radiant flux is equal to half of the total spectral radiant flux maximum value. Accurate determination of bandwidth requires good sampling or interpolation of the measured spectral distribution.
- **Radiant efficiency (of a source)** (see **Section 6.1.2.**)

6.1.4 Auxiliary Source. A sphere with either a radiometer or a spectroradiometer shall be equipped with an auxiliary source. Self-absorption measurements shall be carried out and a correction made for each product under test, unless the correction factors are found to be negligible (much smaller than the desired uncertainty) for the type of products to be tested. The emission from the auxiliary lamp should cover the wavelength range to be measured. In the case of a sphere radiometer, the auxiliary lamp should be an LED with a peak wavelength within 5 nm of that of the DUT, or a source with a spectral distribution similar to that of the DUT (e.g., the lamp used for instrument calibration; see **Section 6.1.2.**). For spectroradiometers, a xenon or deuterium lamp, or a tungsten-halogen lamp operating at a high correlated color temperature (at least 3200 K) in addition to an LED can be used, which could be the source used for the instrument calibration (see **Section 6.1.2.**).

6.1.5 Fluorescence and Phosphorescence. Both fluorescence and phosphorescence cause errors in the measurement of total spectral radiant flux. UV photons emitted by the source are absorbed and re-emitted at a longer wavelength, thus reducing the DUT signal at the absorption wavelengths and increasing the signal at the wavelengths of re-emission. If the calibration source has the same spectral shape as the DUT, the errors will cancel as long as the fluorescence properties of the system are stable, which will not necessarily be the case (for example, fluorescence can change with time as well with as radiation exposure). For spectroradiometers, another solution for correcting fluorescence and phosphorescence errors is to create a fluorescence (stray light) correction matrix by measuring the response of the sphere-spectroradiometer system to individual wavelengths of light.⁴

At a minimum, the radiometer or spectroradiometer system should be characterized for fluorescence and phosphorescence errors by irradiating the sphere with a narrowband source with a peak wavelength within 5 nm of the DUT's peak wavelength and which has less than 0.01% out-of-band light. The measured out-of-band light should be less than 3%.

6.1.6 Stray Light. Stray light is not usually a concern for integrating sphere measurements. However, care should be taken to minimize the external light entering the sphere—for example, around DUTs mounted in a 2 π configuration (per ANSI/IES LM-78-20; see **Section 2.3**).

6.2 Goniometer

A goniometer shall be installed in a dark room or enclosure with low reflectance of UV wavelengths. The room should be temperature controlled and not subject to heat accumulation from the DUT (see **Section 4.1**). Care shall be taken to prevent drafts from ventilation that might affect measurement of DUTs that are temperature sensitive (see **Section 4.2**). ANSI/IES LM-75-19 (see **Section 2.2**) provides general recommendations and requirements for making measurements with goniophotometers.

6.2.1 Detector. The input optics at the detector shall have an approximate cosine response. The cosine response of the detector should have a directional

response index of less than 2% in the wavelength region of interest within the field of view.¹

6.2.2 Radiometer. The radiometer is a spectrally integrating detector that may have a spectral filter to limit the wavelength range of the detection (e.g., to avoid detection of ambient radiation or undesired out-of-band radiation). (See **Section 6.1.2** for radiometer linearity details.)

Calibration: The radiometer shall be calibrated for irradiance traceable to the International System of units (SI) through a national metrology institute (NMI). The calibration method shall take into consideration the spectral output and uncertainty of the calibration source and/or method compared to those of the DUT.

Measured quantities:

- **Radiant intensity distribution:** The radiant flux per unit solid angle in a given direction. For a detector calibrated for irradiance, the radiant intensity distribution, $I(\theta, \phi)$, is the irradiance measured by the radiometer multiplied by the square of the distance from the DUT radiant surface to the detector.
- **Total radiant flux, Φ_e :** The radiant intensity of a source, integrated over all directions (4π sr):

$$\Phi_e = \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} I(\theta, \phi) \sin\theta \, d\theta \, d\phi$$

- **Radiant efficiency (of a source):** The ratio of the total radiant flux, Φ_e , of the emitted radiation to the power, P , consumed by the source. (Source: ANSI/IES LS-1-21; see **Section 2.4**.) The electrical input power P of the DUT shall be measured according to the methodology described in **Section 8**. The radiant efficiency is given as:

$$\eta = \frac{\Phi_e}{P}, \text{ expressed as a percentage}$$

6.2.3 Spectroradiometer. The spectroradiometer shall cover the wavelength range that is relevant for the application, and the bandwidth (full width at half maximum) and scanning interval shall be no greater than 3 nm. Typical systems cover from 200 nm to more than 400 nm. The spectroradiometer system shall

have a wavelength uncertainty within 0.5 nm ($k = 2$). The spectroradiometer system should account for light outside of this wavelength range that may result in stray light within the spectroradiometer system, especially during calibration.

The linearity of an array spectroradiometer should be tested for measurement of such continuous pulses. (per ANSI/IES LM-58-20; see **Section 2.1**.)

Calibration: The spectroradiometer shall be calibrated for spectral irradiance responsivity traceable to the International System of units (SI) through a national metrology institute (NMI).

Measured quantities:

- **Spectral radiant intensity distribution, $I_\lambda(\theta, \phi)$:**
The radiant flux per unit wavelength interval per unit solid angle for a given direction. (Source: ANSI/IES LS-1-21; see **Section 2.4**.) For a spectroradiometer calibrated for spectral irradiance placed at a sufficiently large distance from the DUT (refer to ANSI/IES LM-75-19; see **Section 2.2**), the spectral radiant intensity distribution of the source is related to the measured spectral irradiance, E , by $I = E \cdot d^2$, where d is the distance from the radiant surface of the DUT to the detector.
- **Radiant intensity distribution, $I(\theta, \phi)$:** The spectral radiant intensity of a source, integrated over the wavelength range:

$$I(\theta, \phi) = \int_\lambda I_\lambda(\theta, \phi) d\lambda$$

- **Total radiant flux, $\Phi_e(\lambda)$:** The total radiant intensity of a source, integrated over all directions (4π sr):

$$\Phi_e = \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} I(\theta, \phi) \sin\theta \, d\theta \, d\phi$$

- **Other quantities:** Information on calculation of additional quantities may be found in **Section 6.2.2**.

6.2.4 Stray Light. For goniometer measurements, stray light should be suppressed in the test environment through adequate use of low-reflectance finishes on surfaces, shielding, and baffling. (Note: The UV

reflectance of materials, particularly of black paint or black fabric, may be larger than the reflectance in the visible range; i.e., black may not be as “black” under UV radiation as under visible radiation. In addition, stray light may be measured and subtracted from the UV LED source measurement. (Refer to IES LM-75-19 for more detailed information.)

6.2.5 Angular Resolution and Sampling. The angular resolution shall be fine enough to accurately characterize the DUT. For typical wide-angle, smooth-intensity distributions, a 22.5° lateral (rotation perpendicular to the DUT optical axis) and 2.5° longitudinal (rotation around the DUT optical axis) sampling grid is generally sufficient. Finer angle resolution (smaller test increments) should be used for cases in which the radiant intensity from the DUT is changing rapidly as a function of angle, such as in beam-forming sources. ANSI/IES LM-75-19 provides additional guidance (see **Section 2.2**).

6.2.6 Angular Range. The range of the angular scan shall cover the entire solid angle into which the DUT emits light. Goniometers inherently have an angular region in which light from the DUT is blocked by the mounting hardware. This “dead” angle should be minimized to a point where its effect on total (spectral) radiant flux measurements becomes negligible or appropriate correction procedures are applied.

6.2.7 Goniometer Alignment. The goniophotometer axis (the rotation axis of the UV LED holder) and the position of the radiometer should be accurately co-aligned. This should be checked periodically, as the goniometer axis can drift over time if the mirror angle deflects. Even a small deflection of the mirror can cause large errors in the measurement of narrow-beam DUTs.

The radiometric center of the DUT shall be located such that it coincides with the intersection of the goniometer axes. A description of the location of the radiometric center of the DUT and a description of the orientation of the DUT with respect to the goniometer axes shall be reported. Measurement errors due to goniometer misalignment shall be reduced such that they do not contribute significantly to the total measurement uncertainty.

7.0 Test Preparation

7.1 DUT Identification

It is always good laboratory practice to mark or clearly identify DUTs.

7.2 DUT Handling

While UV LED sources are not as sensitive to movement as incandescent lamps, vibrations and mechanical shocks should be minimized. Devices to be tested should not be stored under temperature extremes or in high humidity conditions.

7.3 Seasoning

Seasoning is not required for the methods described in this document. If the UV LED source is to be measured for rating purposes, it should be seasoned for 100 to 1,000 hours under constant forward current at the T_j used for rating.

7.4 Temperature Stabilization

The T_j shall be stabilized before measurement of the forward voltage, U_F .

7.5 Operating Position and Orientation

The light emission process of an LED is not affected by its orientation. The effect, if any, of thermal convection due to the operating orientation of the DUT is considered negligible with the method described in this document when the DUT's T_j is kept constant.

8.0 Measurement Methods

The procedures described in **Sections 8.1** through **8.5**, depending on the circumstances, shall be used to make optical and electrical measurements. Three measurement methods are described:

- Differential continuous pulse (DCP) method (see **Section 8.2**)
- Mean differential continuous pulse (M-DCP) method (see **Section 8.3**)
- DC or long pulse (DC) method (see **Section 8.4**)
- Continuous pulse time resolved (CPT) method (see **Section 8.5**)

The differential methods or the CPT method, with the pulse parameters listed in **Table 8-1**, shall be used for measurements at or above 1% of the DUT's nominal current (i.e., greater than 10 mA for a 1-amp LED). The DC or long-pulse method should be used for measurements below 1% of the DUT's nominal current. The DC method may also be used for goniometric measurements above 1% of the DUT's nominal current if the average rise in junction temperature during the measurement is offset by a TCM adjustment as described in **Section 8.2**. DC or long-pulse operation provides more signal for the optical detector, increasing the signal-to-noise ratio. **Table 8-1** shows the applicability of each method and its key parameters.

Table 8-1. Measurement Method Parameters and Applicability

Test Current, I_T	Method	Pulse 1 Parameters	Pulse 2 Parameters
$I_T > 10\%$ of I_{nominal}	CPT, DCP or M-DCP*	10 μs , 1% duty cycle	20 μs , 2% duty cycle
$1\% < I_T < 10\%$ of I_{nominal}	CPT, DCP, or M-DCP*	25 μs , 10% duty cycle	50 μs , 20% duty cycle
$I_T < 1\%$ of I_{nominal}	Any method allowed; DC or long pulse should be used	N.A.	N.A.

* DC or long pulse may also be used, only for goniometric measurements, if T_j rise is offset by TCM adjustment.

The two differential measurement methods are built upon a well-documented LED measurement technique: continuous pulse (CP) measurement. The CP measurement technique is described in **Section 8.1**, and four other measurement methods are described in **Sections 8.2** through **8.5**.

8.1 Continuous Pulse Measurement Technique – Foundation for the Differential Methods

8.1.1 Overview. In continuous-pulse (CP) measurements, the DUT is powered with a continuous, low-duty-cycle

train of narrow current pulses. The optical measurement is made for the time-averaged signal. **Figure 8-1** shows an example of current pulses applied and the sequence for CP measurements. D_w , equal to t_{on} , is the width of each current pulse, and D_e , the sum of t_{on} and t_{off} , is the period of repetition of the pulse; the duty cycle is given by D_w/D_e .

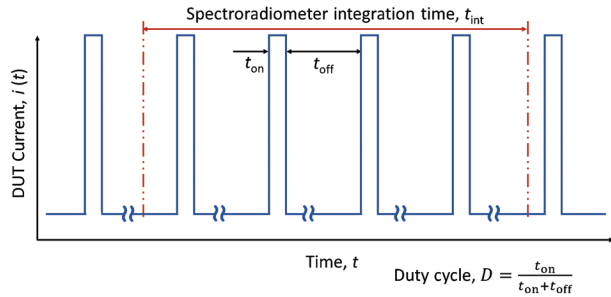


Figure 8-1. An example of a current pulse train in the continuous-pulse technique.

8.1.2 Notes on Instrumentation. The CP technique is suitable for a wide range of measurement devices. However, in this technique, because the duty cycle of the pulse is very low, the detectors of optical measurement instruments and other devices in the signal chain are exposed to very high peaks compared to the average signal level. For example, if the duty cycle is 1%, the peak signal level is 100 times higher than the time-averaged signal level.

When using the continuous-pulse measurement technique, care should be taken to ensure that all instruments and components in the signal chain (e.g., broadband detectors, amplifiers, multimeters) are not saturated with the DUT's high peak emission. For example, a typical silicon photodiode has a nonlinear response above approximately 100 μA of photocurrent output. Therefore, if a photometer head using a silicon photodiode is used to measure a 1% duty cycle CP signal, the time-averaged photodiode current should be 1 μA or less to ensure that the photodiode does not saturate.⁵

Amplifier linearity over the required 100:1 range is also important. This can be tested by connecting a signal generator and comparing the amplifier's output with two different signals: 1) a steady DC current of 1 μA ; and

2) rectangular 100- μA current pulses with a 1% duty cycle. The time-averaged reading of both signals should be the same if the amplifier works properly.

Instruments that auto-range should not be used for continuous-pulse measurements. For example, auto-ranging illuminance meters may not work properly for pulsed light exhibiting a 100:1 dynamic range. Similarly, auto-ranging DMMs should be programmed for a fixed range when making measurements of a continuous-pulse signal. If the continuous-pulse measurements are made using short pulse durations (less than 20 μs) and a low duty cycle, heating during the pulse, as well as average heating, is minimal with a proper heat sink. Therefore, measurement errors due to temperature rise are generally negligible. However, since the optical measurement includes the entire current waveform, imperfections in the current waveform do result in measurement errors. For example, a pulse width error of 2 μs when testing with pulses of 20 μs represents a 10% error.

The primary continuous-pulse error sources are:

- *Pulse width and frequency imperfections:* Duty cycle factor is incorrect.
- *Pulse amplitude distortions:* Tilt, sag, overshoot, and settling time cause assumed average current amplitude to be incorrect.
- *Pulse rise and fall inclusion:* Due to increased quantum efficiency, more light is produced when the DUT is driven at less than the nominal current level. This error is generally positive.
- *Large dark signal:* The detector is measuring the dark signal 99% of the time on a 1% duty cycle. The dark signal can easily become comparable to the main signal.

8.2 Differential Continuous Pulse Method

The differential continuous pulse (DCP) measurement method eliminates the primary continuous-pulse errors by taking the difference of two separate continuous-pulse measurements. The method is shown graphically in **Figure 8-2**. A schematic of the measurement flow is shown graphically in **Figure 8-3**.

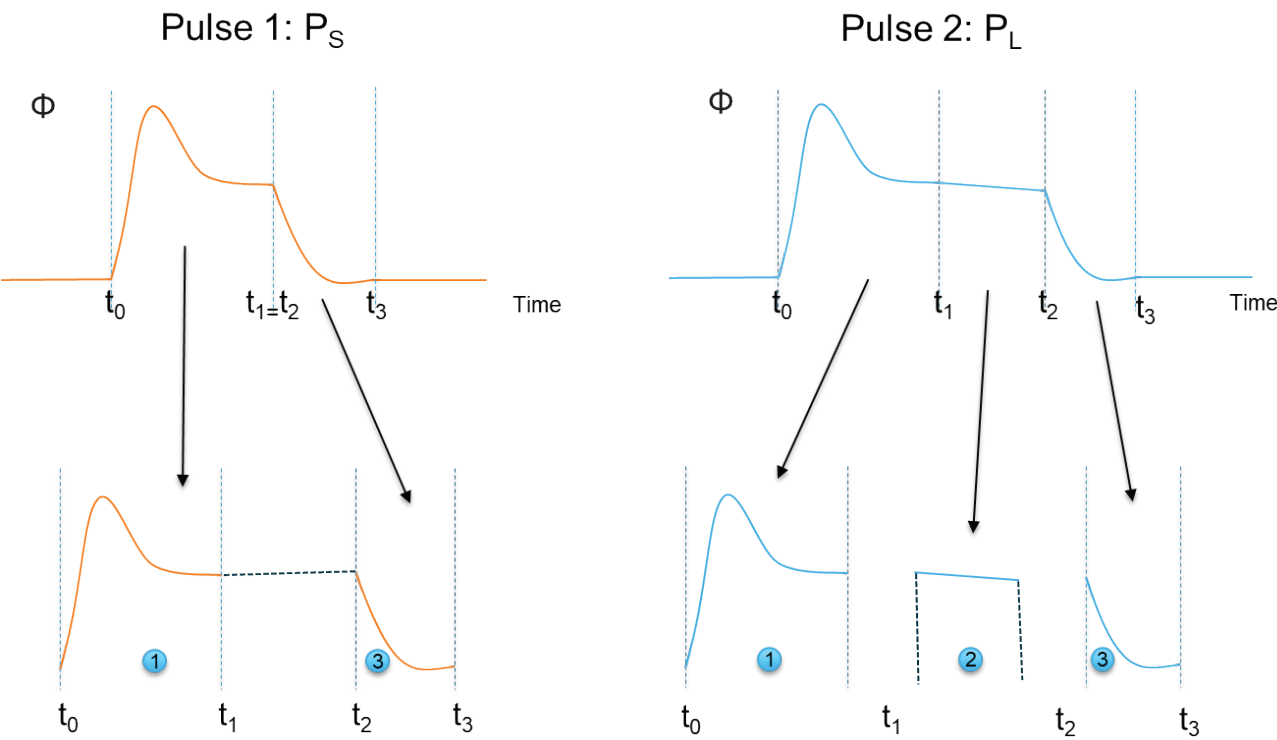


Figure 8-2. Illustration of pulse breakdown in the differential continuous pulse (DCP) method.

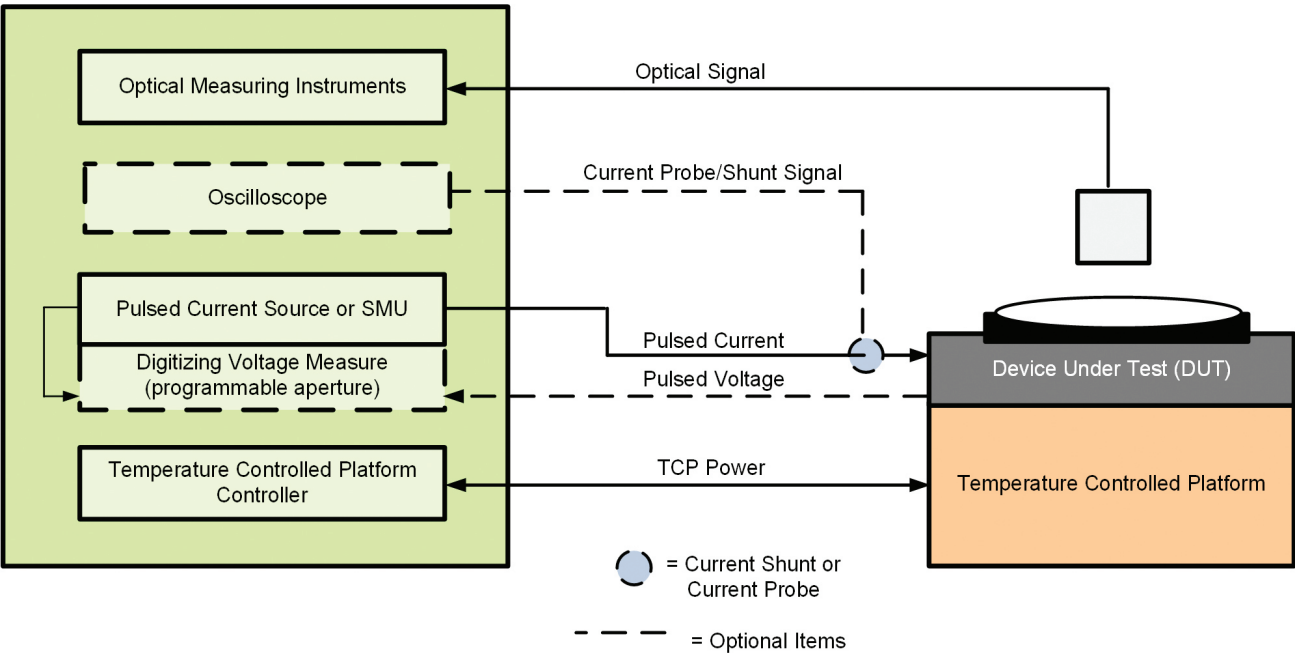


Figure 8-3. Instrument connections for differential continuous pulse measurements.

Two continuous-pulse measurements are used in the DCP method. The first measurement utilizes a shorter pulse (e.g., 10 μ s, 25 μ s) than the second measurement (e.g., 20 μ s, 50 μ s). Both measurements use the same frequency.

A pulse is composed of three regions:

- Rising and settling
- Stable operation
- Falling

For the majority of accurate pulsed-current sources, the rising and settling characteristic is the same regardless of pulse width, P_L . The same is true for the falling characteristic. Therefore, when the second continuous-pulse measurement is subtracted from the first, the contributions from the rising and settling region and the falling region subtract out. The remaining signal is from the second pulse's middle region, stable operation. In this way, the DCP method removes pulse amplitude distortions.

Pulse width imperfections are sometimes similar for two pulses generated by the same PCS. For example, if the source has an error of +1 μ s at 20 μ s, it may exhibit the same error at 10 μ s. If this is the case, then pulse width errors subtract out.

Pulse frequency imperfections may or may not be similar for the first and second measurements. Even if they are similar, the error is not removed by the subtraction.

Most of the error due to the dark signal (the portion corresponding to the off time of the first measurement) is removed by the subtraction.

Thus, the DCP method removes many of the errors associated with the traditional CP technique.

For integrated optical measurements, such as radiant flux and luminous flux measurements, acceptable results can be obtained by subtracting the measured flux values from the pulse measurements. However, for non-integrated measurements, such as peak wavelength, direct subtraction is not valid. For these non-integrated

measurements, the respective short pulse (P_S) and long pulse (P_L) spectral power distributions (SPDs) shall be subtracted first, and then the parameter of interest shall be calculated from the resulting difference SPD.

The principle of the DCP method is described in the equations that follow (refer also to **Figure 8-4**, following Step 5). The flux generated by the short pulse, P_S , can be decomposed into three components:

$$\begin{aligned}\Phi_{P_S} &= \frac{1}{(t_3 - t_0)} \int_{t_0}^{t_3} \Phi_t(t) dt \\ &= \Phi_{P_S, t_0 \rightarrow t_1} + \Phi_{P_S, t_1 \rightarrow t_2} + \Phi_{P_S, t_2 \rightarrow t_3}\end{aligned}$$

For the short pulse, $t_1 = t_2$. Thus, for the second component, $\Phi_{P_S, t_1 \rightarrow t_2} \equiv 0$

The flux generated by the long pulse, P_L , can also be decomposed into three components:

$$\begin{aligned}\Phi_{P_L} &= \frac{1}{(t_3 - t_0)} \int_{t_0}^{t_3} \Phi_t(t) dt \\ &= \Phi_{P_L, t_0 \rightarrow t_1} + \Phi_{P_L, t_1 \rightarrow t_2} + \Phi_{P_L, t_2 \rightarrow t_3}\end{aligned}$$

Assuming that $\Phi_{P_L, t_0 \rightarrow t_1} = \Phi_{P_S, t_0 \rightarrow t_1}$, the measured DCP result is:

$$\Phi_{P_L} - \Phi_{P_S} = \Phi_{P_L, t_1 \rightarrow t_2} + [\Phi_{P_L, t_2 \rightarrow t_3} - \Phi_{P_S, t_2 \rightarrow t_3}]$$

The first term resulting from the subtraction, $\Phi_{P_L, t_1 \rightarrow t_2}$, represents a trapezoidal area with vertical sides and a sloped top.

The second term, $[\Phi_{P_L, t_2 \rightarrow t_3} - \Phi_{P_S, t_2 \rightarrow t_3}]$, is a negligible value that is actually negative because the P_L flux output at t_2 is less than the P_S flux output at t_2 .

Thus, the result of the subtraction is approximately the middle component of the long pulse:

$$\Phi_{P_L} - \Phi_{P_S} \approx \Phi_{P_L, t_1 \rightarrow t_2}$$

The following steps 1 through 5 shall be followed to make DCP measurements:

Step 1: Stabilize the temperature

When using the ambient temperature control method: For the room temperature condition or elevated temperature condition using a thermal chamber, the ambient temperature or thermal chamber air temperature shall be set to the desired T_J . If the DUT exhibits a stable U_F characteristic, stabilization of the DUT's T_J may be monitored by applying a small monitoring DC current, I_M , (e.g., 1 mA) through the DUT while monitoring the DC or pulsed voltage, U_M . When the measured U_M is stabilized within a range determined by the DUT's U_F vs. temperature characteristic (k -factor) and the measured parameter's temperature sensitivity factor, the DUT is considered stabilized. If the temperature of the DUT at its thermal contact point (pin, case, or board) is monitored, it is considered stabilized if the monitored temperature remains within $\pm 0.5^\circ\text{C}$ from the specified T_J for at least 5 minutes.

When using a TCM: The DUT is mounted on the TCM, the TCM temperature is set to the specified T_J , and the TCM temperature is allowed to reach and stabilize to the set temperature. The TCM temperature is considered stabilized if it remains within $\pm 0.2^\circ\text{C}$ from the specified T_J for at least 1 minute.

Step 2: Apply the continuous-pulse current (P_S and P_L)

After programming the PCS to generate the required P_S or P_L waveform, wait for the time required for the PCS current output to stabilize. For a PCS with automatic pulse width correction, the wait time should be greater than 20 pulses to allow the correction to occur. If shorter wait times are desired, automatic pulse width correction should be disabled.

Step 3: Perform the optical and electrical measurements

Perform optical measurements for a specified duration of integration time for the time-averaged signal. If the measurement is not synchronized with the pulse frequency, the integration time shall be exactly an integer multiple of the period of the pulses to avoid errors due to the timing of the integration.

Step 4: Perform the subtraction and scale the results

Subtract flux values or subtract spectrums to obtain the

raw DCP measurements. Scale the resulting spectrum or measured photometric or radiometric values using:

$$(\text{scaled values, } y_i) = (\text{measured values})(\text{integration time}) / (\text{total "on" time during integration})$$

Step 5: Measure the DUT U_F

The DUT U_F is an important parameter that should be measured whenever an optical measurement is made. For visible-range LEDs and some UV LEDs, U_F is a characteristic primary function of the LED's forward current and its temperature. However, in many UV LEDs, a transient voltage effect distorts the U_F during the first milliseconds of operation. The effect can be seen in **Figure 8-4**; the distortion can be greater than 1 V for some LEDs.

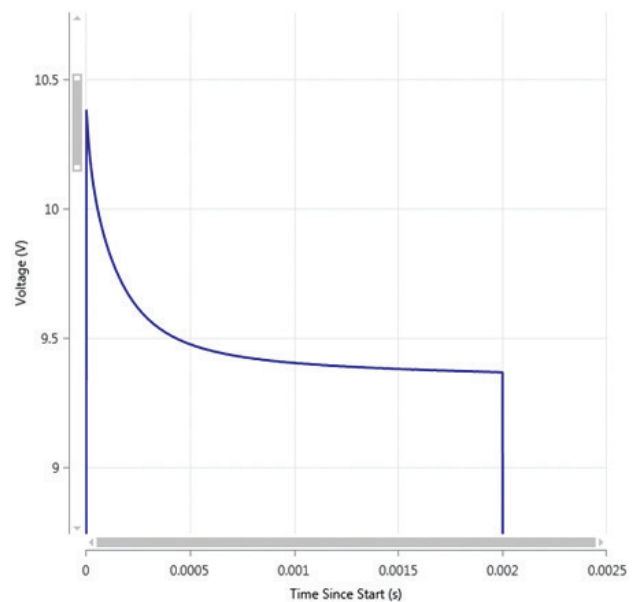


Figure 8-4. UV-C LED U_F vs. time for a 2-ms, 200-mA current pulse, showing a 900-mV transient voltage effect. (Data collected with Vektrex SMU digitizer)

For these LEDs, U_F shall be measured after a delay sufficient for the transient voltage effect to dissipate. (Refer to **Annex B** for a method for determining whether an LED has a transient voltage effect.)

For DUTs with no transient voltage effect, the average U_F should be measured during the second half of the DCP long pulse. This shall be accomplished using a DMM (or a digitizer) that is triggered with a hardware trigger signal from the PCS. This hardware triggering ensures that

the average measurement corresponds to the optical measurement time (t_1 to t_2 of pulse P_L). The DMM shall be configured to have a trigger delay time (t_0 to t_1 of pulse P_L) and a measurement aperture time that matches the optical measurement time (t_1 to t_2 of pulse P_L).

For DUTs with a transient voltage effect, the U_F shall be measured after a delay. However, during the delay the DUT's junction heats up, and thus the external TCM temperature can no longer be assumed equal to the T_J . To correct for this, the TCM temperature shall be reduced by an amount equal to the T_J rise during the delay. Since U_F is distorted by the transient voltage effect during the delay, the DUT's optical output shall be used to estimate the junction temperature rise.

The U_F may be measured using a DC or long-pulse measurement method. The DC method is simpler; however, it may require a TCM temperature that is below the ambient dew point, resulting in a risk of condensation.

DC U_F measurement method:

- Mount the DUT to a TCM, power it with DC current at the specified I_F , and measure the optical output (e.g., radiant flux, peak wavelength).
- Lower the TCM temperature until the measured optical value matches the value y_i obtained in Step 4 above.
- Record the U_F under this DC operation, which is the true U_F at the specified T_J under DCP.

Long-pulse U_F measurement method (refer to Figure 8-5):

- Mount the DUT to a TCM, power it with pulsed current at the specified I_F , and measure the optical output (e.g., radiant flux, peak wavelength). Delay the onset of the optical measurement until the transient voltage effect has dissipated.
- Lower the TCM temperature by a fixed amount (e.g., the approximate T_J rise), and measure the optical output again.
- Calculate the temperature sensitivity factor for the measured value over the integration time as:

$$[y_i(2) - y_i(1)] / [(TCM \text{ Temperature } 2) - (TCM \text{ Temperature } 1)]$$

- Using this sensitivity factor, calculate a third TCM temperature that will result in an optical measurement that equals the DCP measurement.
- Adjust the TCM to this third temperature. Confirm that the pulsed optical measurement matches the DCP measurement within acceptable uncertainty.
- Measure the average DUT U_F under this long-pulse operation. Ensure that the voltage measurement aperture corresponds exactly to the beginning and end of the effective optical integration time, including any detector line binning time.
- Record the U_F under this condition, which is the true U_F at the specified T_J under DCP.

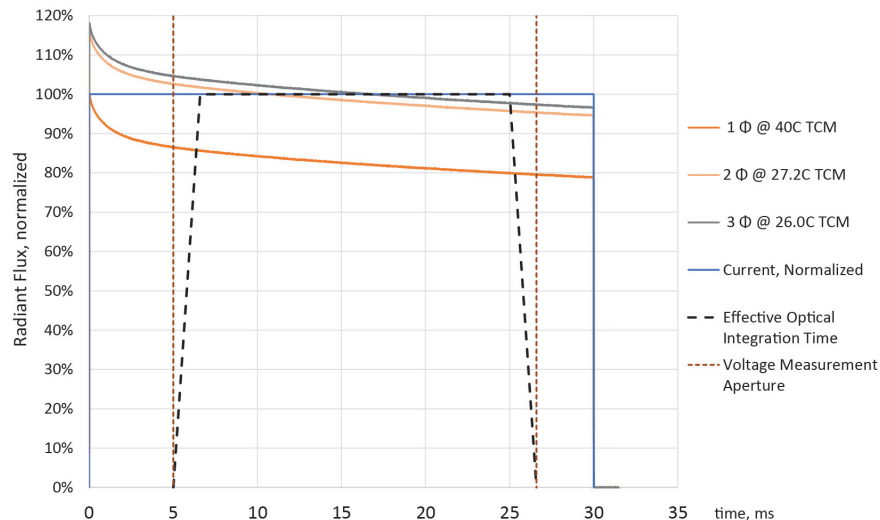


Figure 8-5. Long-pulse U_F measurement method using a 30-ms pulse.

8.3 Mean Differential Continuous Pulse (M-DCP) Method

The M-DCP method improves on the DCP method by adding a correction for the following errors that are not corrected in the DCP method:

- Pulse width errors
- Pulse frequency errors
- Pulse amplitude distortions during the stable part of the pulse

These errors are corrected by measuring the mean current and/or signal amplitude for both the P_S and the P_L waveforms, using a DMM with a multi-slope integrator. These mean currents are then used to create a correction factor that is applied to the DCP results. The M-DCP method should be selected whenever the measurement uncertainty due to PCS timing or amplitude errors needs to be minimized or reduced. For example, **Table 8-2** illustrates a typical uncertainty

calculation for a DCP measurement of 10 μs (1%) and 20 μs (2%).

Table 8-2. Calculation of PCS Errors

PCS Characteristic	Specified Accuracy	Contribution to Uncertainty
Pulse width accuracy, P_S	0.2 μs	1% (see note)
Pulse width accuracy, P_L	0.2 μs	0.5% (see note)
Pulse width jitter, P_S	0.11 μs	1.1%
Pulse width jitter, P_L	0.11 μs	0.55%
Pulse frequency jitter, P_S	2 μs	0.2%
Pulse frequency jitter, P_L	2 μs	0.2%
Root Sum Squared Total:		1.69%

Table note: Uncertainty calculation assumes half of PCS pulse width error is common to both P_S and P_L .

A schematic workflow for the M-DCP measurements is shown in **Figure 8-6**.

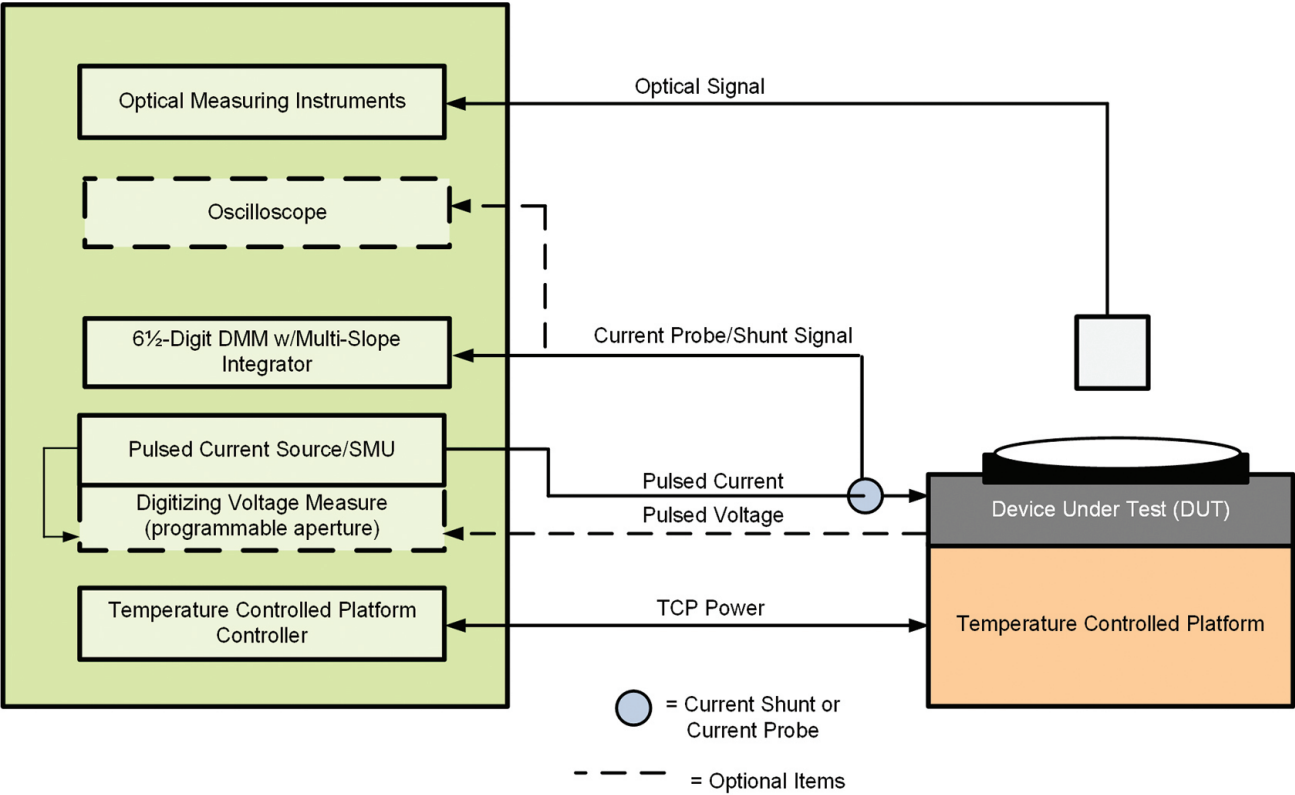


Figure 8-6. Instrument connections in the mean differential continuous pulse (M-DCP) method.

The following steps 1 through 5 shall be followed to make M-DCP measurements:

Step 1: Stabilize the temperature

Perform temperature stabilization as in the DCP method.

Step 2: Apply the continuous current pulse P_S

Apply the short pulsed current waveform as described in the DCP method.

Step 3: Perform optical and electrical measurement

Measure the optical and electrical mean values under the P_S condition, $Y_{i,P_S,mean}$. Also measure the mean current value of the P_S waveform using the DMM with multi-slope integrator or high-speed DAQ.

Note: Some higher-end DMMs (7½ or 8½ digits) may have some issues that result in a slightly higher or lower measurement result. This error cancels out when the differential method is used. However, it is still recommended that the DMM be checked to ensure it is working properly with the high frequency content of the signal by changing the DMM to a range that is one level higher than it is supposed to be. If the measurement results using both ranges are similar, the DMM is working properly.

Step 4: Repeat Step 2 and Step 3 for the P_L condition

Obtain the optical and electrical measurement mean values under the PL long-pulse condition, $Y_{i,P_L,mean}$.

Step 5: Calculate the mean values

Calculate the differential mean values using:

$$Y_{i,mean} = Y_{i,P_L,mean} - Y_{i,P_S,mean}$$

Step 6: Convert to DC operation values

Convert the differential mean value, $Y_{i,mean}$, to the value Y_i when the DUT is operated under the specified DC current, I_F , using:

$$Y_i = Y_{i,mean} \times \frac{I_F}{I_{F,mean}}$$

Step 7: Measure the DUT U_F

Measure the DUT U_F as in the DCP method (see **Section 8.2**).

8.4 DC Method

The DC method may be used if the measurement current is less than 1% of the DUT's nominal current. In this case, heating errors are negligible, so it can be used for low-current DUT characterization. For DC measurements, the required current shall be applied to the DUT, and electrical and optical measurements shall be made after a suitable delay, long enough for the current to stabilize and any transient voltage effect to dissipate.

The DC method may also be used for goniometric measurements at currents above 1% of the DUT's nominal current if the average junction-temperature rise during the measurement is offset by a TCM adjustment. The long-pulse U_F measurement method procedure described in **Section 8.2** shall be used for determining the offset.

8.5 Continuous Pulse Time Resolved (CPT) Method

The optical quantities of interest can be calculated from subsets of the data best representing the steady state behavior of the DUT (see **Figure 8-7**). The optical detection system is required to be fast in order to allow a dense sampling of the optical pulse signal. In other words, the larger of rise-time and fall-time of the detector shall be at most 10% of the width of the current pulse driving the DUT (e.g., 1 µs for a 10-µs pulse).

The CPT method cannot be used to make spectral measurements. The CPT method eliminates timing-

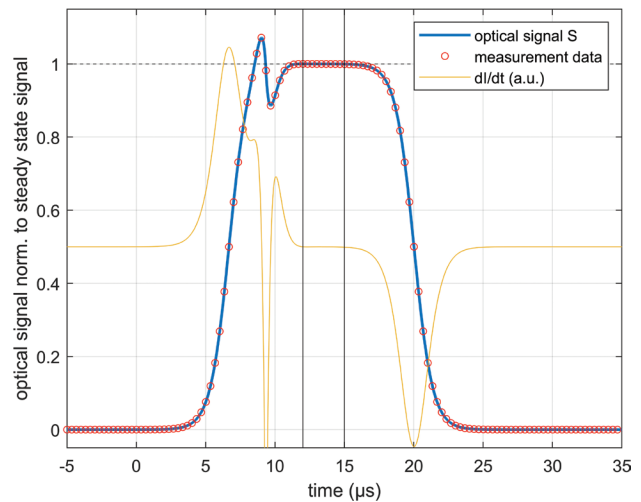


Figure 8-7. Example of a CPT measurement of a pulsed optical signal.

related sources of error by selecting a suitable subset of optical data corresponding to a flat portion of the current pulse. Detector dark signal offsets can also be determined directly from the optical data recorded during the pulse “off” time.

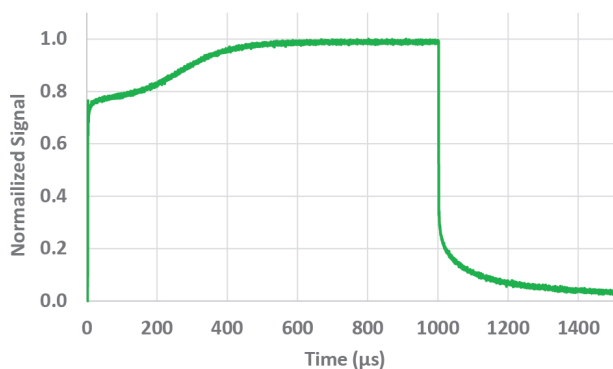
Figure 8-7 shows an example CPT measurement. The blue line is the optical signal; red circles indicate measurements with a fast detector (rise and fall time much less than pulse width) and good sampling. Black vertical lines indicate time interval (subset of data) used for measurement of steady-state values of optical quantities identified by a plateau in the first derivative of the current signal (yellow line). Due to transient heating, the optical signal is not flat. The plateau should be identified based on the current pulse.

Certain commercial photodiodes exhibit a capacitance when exposed and shuttered to optical radiation. This capacitance causes a time-dependent signal, which is not acceptable for time-resolved measurements. **Figure 8-8** shows the response curve of two photodiodes with respect to time. The first photodiode has a charging effect that occurs over 500 μs and a discharging effect lasting over 500 μs . The second photodiode has a much faster response time. Photodiodes should be selected for which settling times are much shorter than the width of the applied current pulse. This is not a problem for the M-DCP method.

The following steps 1 through 5 shall be followed to make CPT measurements:

Step 1: Stabilize the temperature

Perform temperature stabilization as in the DCP method.



Step 2: Apply the continuous current pulse, P_s

Apply the short pulsed current waveform as described in the DCP method.

Step 3: Perform optical and electrical measurement

Sample the optical and electrical signals. Determine the time interval to be used for the optical measurement by analyzing the first derivative of the sampled current signal.

Step 4: Measure the DUT U_F

Measure the DUT U_F as in the DCP method (see **Section 8.2**).

9.0 Measurement Uncertainty

If a statement of uncertainty is required, the recommendations given in ISO/IEC Guide 98-3:2008 should be followed.⁶ For all measurements, expanded uncertainty with a confidence interval of 95% shall be used, thus, in most cases, using the coverage factor $k = 2$.

To verify the measurement uncertainty, the measurement results of some DUTs may be compared with measurements traceable to an NMI.

10.0 Test Report

The report shall list all pertinent data concerning

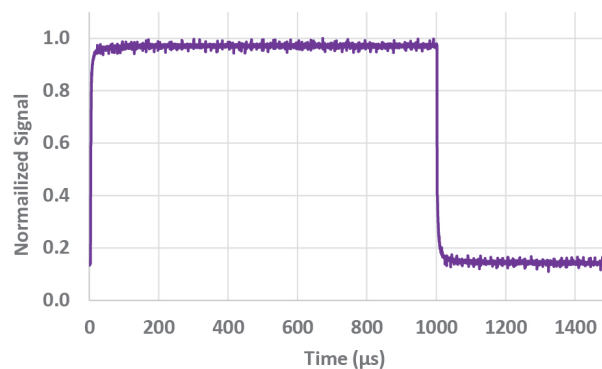


Figure 8-8. Signal-response curves of two photodiodes with respect to time.

conditions of testing. The report shall also list type of equipment, the DUT, and reference standards used.

Typical items reported are:

- Date of measurement and testing agency
- Manufacturer's name and designation of the DUT
- Measured quantities: e.g., total radiant flux, radiant efficiency, peak wavelength, centroid wavelength
- Rated electrical values: electrical input power and forward current
- Method of measurement details: pulse width and duty cycle
- Junction temperature of the DUT and ambient temperature
- Instrument used: sphere-spectroradiometer, sphere-detector, gonio-detector, or gonio-spectroradiometer, and 2π or 4π geometry
- Designation and type of reference standard used: 2π or 4π , spectral radiant intensity or spectral irradiance, standard source or standard LEDs, peak wavelength for LED standard
- Correction factors applied: e.g., self-absorption
- Optical measurement conditions: diameter of sphere, goniometer test distance
- Measurement parameters: input current, forward voltage, power
- Spectral power distribution, if applicable
- Bandwidth of spectroradiometer, if spectral distribution is reported
- Other equipment used
- Statement of uncertainties, if required
- Deviation from standard operating procedures, if any

Annex A – Determination of Temperature Stability

To determine stability, the LED is driven at a low measurement current, ideally less than 0.5% of nominal current.

The LED U_F is monitored until it reaches stability. Stability is achieved when the U_F variation plus the

additional expected U_F change is small enough so that the corresponding error in the measured parameter of interest (for example, flux) is below the acceptable error. The error can be calculated as follows:

1. Divide the residual U_F variation plus the remaining expected U_F settling by the expected k -factor ($\Delta U / \Delta T$), at the measurement current, to obtain the residual temperature variation. (Note: this k -factor can be an average or typical value for the part type.)
2. Multiply the obtained ΔT by the measured parameter's temperature factor (e.g. $\Delta \Phi / \Delta T$) to obtain the expected measured parameter deviation.
3. Divide this expected deviation by the measured parameter's expected value to determine the percent error in the measured parameter. For example: (deviation of 3 lm)/(expected value of 300 lm) = 1% error

When this variation is below the error budget for the starting temperature error, stability is achieved.

Annex B – Determination of Transient voltage Effect

As shown in **Figure B-1**, the transient voltage effect distorts the LED's U_F during the first milliseconds of operation, superimposing a decaying voltage transient on top of the LED's normal voltage-versus-time curve. As is well documented in JESD51-51,⁷ the U_F vs. t curve is normally a function only of the DUT's junction temperature. Thus, the presence of a transient voltage effect can be inferred if the U_F vs. t curve is determined not to correspond to junction temperature.

To determine whether a U_F vs. t curve corresponds to junction temperature, U_F is plotted vs. t for two temperatures, for example, 25 °C and 26 °C. The results are graphed on a square-root-of-time scale. This is illustrated in the following example.

Figure B-1 shows a plot of a UV-C LED's U_F vs. t curve at two case temperatures: 25 °C and 26 °C. Voltage is on the vertical axis, and the square root of time is on the horizontal axis. Also plotted is the instantaneous

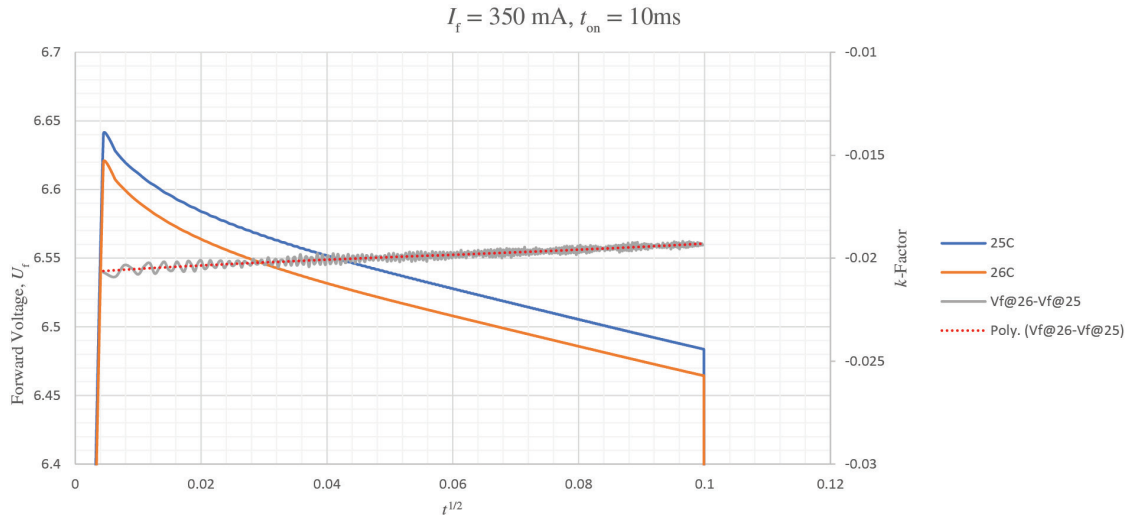


Figure B-1. UV-C LED showing no transient voltage effect and a reasonable calculated T_j rise of 7.75 °C. In the key, “Poly.” refers to “polynomial fit.” (Data collected with Vektrex SMU digitizer)

k -factor, $(U_F \text{ at } 26^\circ\text{C}) - (U_F \text{ at } 25^\circ\text{C})$. Several things can be seen in this plot:

- The curves are approximately linear; this corresponds to a device with a square junction area receiving constant heating.
- The junction rise, calculated from the starting and ending points of the U_F curves and the k -factor, is about 7.75 °C.
- The k -factor plot is relatively linear.

A junction temperature rise of 7.75 °C is reasonable for this LED and its mount. This DUT shows no transient voltage effect.

Figure B-2 shows a plot of a similar UV-C LED's U_F vs. t curve, again at two case temperatures: 25 °C and 26 °C. This plot shows:

- The curves are not linear; this could mean a more complex thermal path, or it could be due to the transient voltage effect.
- The junction rise, calculated from the starting and ending points of the U_F curves and the k -factor, is about 75 °C.
- The k -factor plot is not linear.

A junction temperature rise of 75 °C is not reasonable for this LED and its mount. This DUT shows a transient voltage effect.

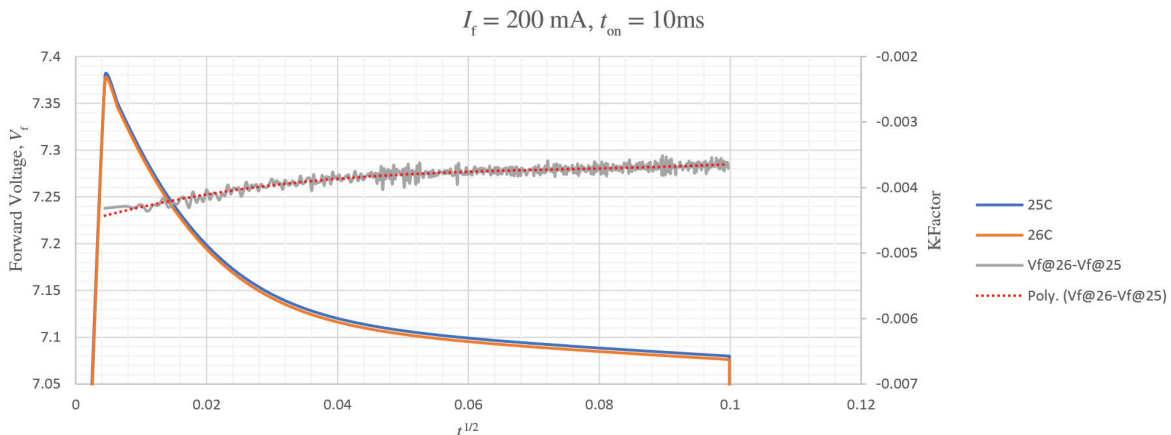


Figure B-2. A UV-C LED with transient voltage effect and an unreasonable calculated T_j rise of 75 °C. In the key, “Poly.” refers to “polynomial fit.” (Data collected with Vektrex SMU digitizer)

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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